

B.TECH.
(SEM V) THEORY EXAMINATION 2022-23
PROCESS DYNAMICS & CONTROL

Time: 3 Hours

Total Marks: 100

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.

2 x 10 = 20

- (a) What is the objective of process control?
- (b) Define interacting system.
- (c) Define MIMO systems.
- (d) Define interacting system. Differentiate between open and closed loops
- (e) Describe the Nyquist stability criteria with important properties.
- (f) Define in physical terms the servo & regular control problems.
- (g) What are the advantages of auto tuning?
- (h) Which controller is known as rate controller?
- (i) Discuss the advantage & disadvantages of Cascade control.
- (j) Explain DCS in brief.

SECTION B

2. Attempt any three of the following:

10 x 3 = 30

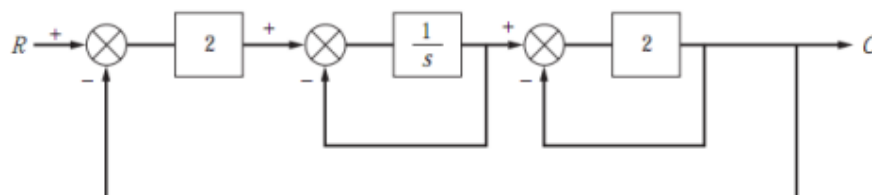
- (a) What are the various forcing functions used to study the dynamic response of a system? Explain with neat sketches.
- (b) Discuss the characteristic of under damped response & plot the graph of overshoot & decay versus damping factor.
- (c) Differentiate between a negative feedback & positive feedback control system with suitable example
- (d) Discuss the term 'feed-forward control'. With the help of a general block diagram, describe the feedback control of a process.
- (e) Discuss the graphical rules of bode diagram.

SECTION C

3. Attempt any one part of the following:

10 x 1 = 10

- (a) Examine the response of 'interacting & non-interacting systems in series' with the help of suitable examples.
- (b) Determine the transfer function of $C(s)/R(s)$ for the control system given below.



4. Attempt any *one* part of the following: 10 x 1 = 10

- (a) Explain online tuning. In online tuning discuss open loop & closed loop method.
- (b) Discuss the Routh's test for testing the stability of a control system. What other information are obtained by Routh's theorem about the control system?

5. Attempt any *one* part of the following: 10 x 1 = 10

- (a) Draw the root locus for a control system having following characteristic equation
 $1 + [\{ k (16s+1) \} / \{ s(s+4)(s+8) \}] = 0$
- (b) Write in brief about tuning rules: i) Ziegler-Nichols tuning rules ii) Cohen – Coon tuning rules

6. Attempt any *one* part of the following: 10 x 1 = 10

- (a) Differentiate between servo control problem and regulator control problem with the help of suitable examples?
- (b) Write the procedure for drawing the Root locus plot.

7. Attempt any *one* part of the following: 10 x 1 = 10

- (a) With a neat sketch explain the components of control system.
- (b) Demonstrate the working & application of SCADA & DCS.